

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 5: Cluster Analysis

Ch 5 – Cluster Analysis – Numerical Problems & Solutions

135 numerical marks, 7.0% of the archive, 37% of this chapter · 3 methods · 11 PYQ instances · every distance recomputed. K-means appears in 7 papers and hierarchical dendrograms in 4, so these two methods alone are worth around 60 marks across the archive.

Methods

- **M1** K-means 78 Bh, 74 Ash, 75 Ash, 71 Shr, 76 Ch, 81 Ba
- **M2** Agglomerative hierarchical clustering + dendrogram 80 Bh, 80 Ba, 76 Ch
- **M3** DBSCAN point classification *practice*

Every problem below ends with its solution drawn step by step. Each panel carries a chip naming the move it makes; the decisive step is framed **KEY STEP**. The table says what each move is and where it is used.

The moves every chapter-5 numerical below is built from

Move	What it says	Where it is used (problem, step; orange = the key step)
seed rule	State the seeds; with one given, take the point farthest from it.	1.1/1.2 1; 1.3 1; 1.4 1; 1.7 1; 1.8 1
assign nearest	Each point joins the centroid it is nearest to.	1.1/1.2 2, 3 ; 1.3 2, 3 ; 1.4 2, 3 ; 1.5 1,2, 3 ; 1.7 2, 3 ; 1.8 2, 3 ; 1.9 1,2,3,4; 1.10 1,2, 3
tie-break	Equal distances: name a rule and apply it; say which way the point went.	1.4 2
move to mean	Each centroid moves to the mean of its members.	1.1/1.2 2; 1.3 2; 1.4 2 ; 1.5 1,2; 1.7 2; 1.8 2; 1.9 1,2,3; 1.10 1,2, 3
unchanged = done	An iteration that reassigns nothing proves convergence.	1.1/1.2 3 ; 1.3 3 ; 1.4 3; 1.5 3 ; 1.7 3 ; 1.8 3; 1.9 4
Manhattan	$ dx + dy $ instead of the straight line.	1.10 1,2, 3
SSE	Sum of squared distances to each point's own centroid: what K-means minimises.	1.9 5
smallest entry	Merge the two clusters with the smallest distance; that value is the bar height.	2.1 1,2, 3 ,4,5; 2.2 1,2, 3 ,4,5; 2.3a 1,2, 3 ,4,5; 2.3b 1,2, 3 ,4,5
MIN update	Single link: the merged row takes the smaller of the two old rows.	2.1 2, 3 ,4,5; 2.3a 2, 3 ,4,5
MAX update	Complete link: the merged row takes the larger of the two old rows.	2.2 2, 3 ,4,5; 2.3b 2, 3 ,4,5
ϵ-neighbourhood	Count the points within ϵ , the point itself included, $d \leq \epsilon$.	3.1 1; Practice 1
core / noise	Count $\geq \text{MinPts}$ is core; not core and near no core point is noise.	3.1 2 ; Practice 2
connect cores	Cores within ϵ of each other form one cluster; borders attach.	3.1 3; Practice 3

M1. K-means

Method

1. State K and the initial centroids. **Always write down which points you seeded with** — the answer depends on it and the examiner needs to follow you.
 2. Build a distance table: every point against every centroid.
 3. Assign each point to its nearest centroid.
 4. Recompute each centroid as the **mean** of its members.
 5. Repeat until the assignment does not change. **That repetition is the convergence proof.**
- You may compare *squared* distances to avoid square roots — the ordering is identical. Say that you are doing it.

Problems 1.1 and 1.2 2+6 2+8

QUESTION AS PRINTED 2078 Bhadra · Q8 · [2+6]

Write K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	2.5	1
3	3.5	1.5
4	4	1
5	3.5	2.5
6	5	3

QUESTION AS PRINTED 2074 Ashwin · Q7 · [2+8]

Write K-means algorithm and find clusters for following data set.

Instance	X	Y
1	1.0	2.0
2	2.5	1.0
3	3.5	1.5
4	4.0	1.0
5	3.5	2.5
6	5.0	3.0

(Take $K = 2$)

The same 6 instances, only the column names differ.

Seeds

Take the two **most widely separated** points, $c_1 = p_1(1, 2)$ and $c_2 = p_6(5, 3)$. (A standard, defensible heuristic; state it.)

Iteration 1

Point	$d \text{ to } c_1(1, 2)$	$d \text{ to } c_2(5, 3)$	Cluster
$p_1(1, 2)$	0	4.123	C1
$p_2(2.5, 1)$	1.803	3.202	C1
$p_3(3.5, 1.5)$	2.550	2.121	C2
$p_4(4, 1)$	3.162	2.236	C2
$p_5(3.5, 2.5)$	2.550	1.581	C2
$p_6(5, 3)$	4.123	0	C2

- $C1 = \{p_1, p_2\} \rightarrow \text{new centroid } \left(\frac{1+2.5}{2}, \frac{2+1}{2}\right) = (1.75, 1.5)$
- $C2 = \{p_3, p_4, p_5, p_6\} \rightarrow \text{new centroid } \left(\frac{3.5+4+3.5+5}{4}, \frac{1.5+1+2.5+3}{4}\right) = (4, 2)$

Iteration 2

Point	$d \text{ to } (1.75, 1.5)$	$d \text{ to } (4, 2)$	Cluster
p_1	0.901	3.000	C1
p_2	0.901	1.803	C1
p_3	1.750	0.707	C2
p_4	2.305	1.000	C2
p_5	2.016	0.707	C2
p_6	3.580	1.414	C2

- **Assignment unchanged → converged.**

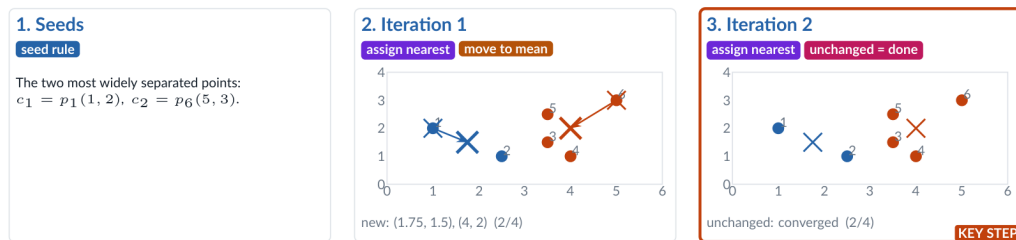
Final answer

- **Cluster 1** = $\{(1, 2), (2.5, 1)\}$, centroid $(1.75, 1.5)$
- **Cluster 2** = $\{(3.5, 1.5), (4, 1), (3.5, 2.5), (5, 3)\}$, centroid $(4, 2)$
- Converged in **2 iterations**.

Seed sensitivity — worth a line, since the question asks for limitations. Seeding instead with the *first* two points, $c_1 = p_1(1, 2)$ and $c_2 = p_2(2.5, 1)$, gives $C1 = \{p_1\}$ and $C2 = \{p_2, p_3, p_4, p_5, p_6\}$, which also converges but leaves a **degenerate 1-point cluster**.

Same data, same K , **different answer** — this is exactly the “sensitive to initial centroids / converges to a local optimum” limitation. **verified/added**

Step by step: seeds, then each iteration (x = centroid; hollow = before, filled = after)



Problem 1.3 4+6

QUESTION AS PRINTED 2075 Ashwin · Q7 · [4+6]

Explain K-means clustering with limitation. Generate two clusters from following dataset using K-means clustering.

A	B
1	2
2.5	4.5
4	6
3.5	4
4	5.5
3	6

Seeds

$c_1 = (1, 2)$ (clearly isolated) and $c_2 = (4, 6)$. (State the choice.)

Iteration 1

Point	d to $c_1(1, 2)$	d to $c_2(4, 6)$	Cl.
(1, 2)	0	5.000	C1
(2.5, 4.5)	2.915	2.121	C2
(4, 6)	5.000	0	C2
(3.5, 4)	3.202	2.062	C2
(4, 5.5)	4.610	0.500	C2
(3, 6)	4.472	1.000	C2

- $C1 = \{(1, 2)\} \rightarrow$ centroid stays **(1, 2)**
- $C2 \rightarrow$ centroid

$$\left(\frac{2.5+4+3.5+4+3}{5}, \frac{4.5+6+4+5.5+6}{5} \right) = (3.4, 5.2)$$

Final answer

$C1 = \{(1, 2)\}$, $C2 = \{(2.5, 4.5), (4, 6), (3.5, 4), (4, 5.5), (3, 6)\}$, centroid $(3.4, 5.2)$.

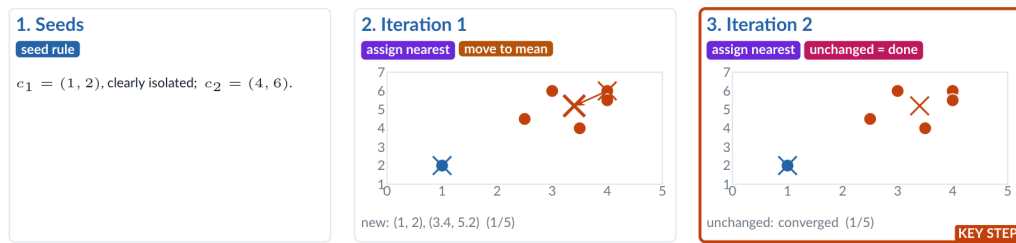
- **Note:** (1, 2) is a genuine outlier here, so the 1-point cluster is the honest answer, not a seeding artefact. Say so, and add that **K-medoids** or **DBSCAN** would treat it as noise.

Iteration 2

Point	d to $(1, 2)$	d to $(3.4, 5.2)$	Cl.
(1, 2)	0	4.000	C1
(2.5, 4.5)	2.915	1.140	C2
(4, 6)	5.000	1.000	C2
(3.5, 4)	3.202	1.204	C2
(4, 5.5)	4.610	0.671	C2
(3, 6)	4.472	0.894	C2

- **Assignment unchanged → converged** after 2 iterations.

Step by step: seeds, then each iteration



Problem 1.4 10

QUESTION AS PRINTED 2071 Shrawan · Q8 · [10]

Explain K-means clustering with limitation Use k-means clustering to cluster the following dataset.

A	B
1.0	1.0
1.5	2.0
3.0	4.0
5.0	7.0
3.5	5.0
4.5	5.0
3.5	4.5

The paper prints no K ; take $K = 2$, the value its sister questions use.

Seeds

 $c_1 = (1, 1)$, $c_2 = (5, 7)$ – the two most widely separated points.

Iteration 1

Point	d to $c_1(1, 1)$	d to $c_2(5, 7)$	Cl.
(1, 1)	0	7.211	C1
(1.5, 2)	1.118	6.104	C1
(3, 4)	3.606	3.606	tie
(5, 7)	7.211	0	C2
(3.5, 5)	4.717	2.500	C2
(4.5, 5)	5.315	2.062	C2
(3.5, 4.5)	4.301	2.915	C2

Iteration 2

Point	d to (1.25, 1.5)	d to (3.9, 5.1)	Cl.
(1, 1)	0.559	5.022	C1
(1.5, 2)	0.559	3.921	C1
(3, 4)	3.052	1.421	C2
(5, 7)	6.657	2.195	C2
(3.5, 5)	4.161	0.412	C2
(4.5, 5)	4.776	0.608	C2
(3.5, 4.5)	3.750	0.721	C2

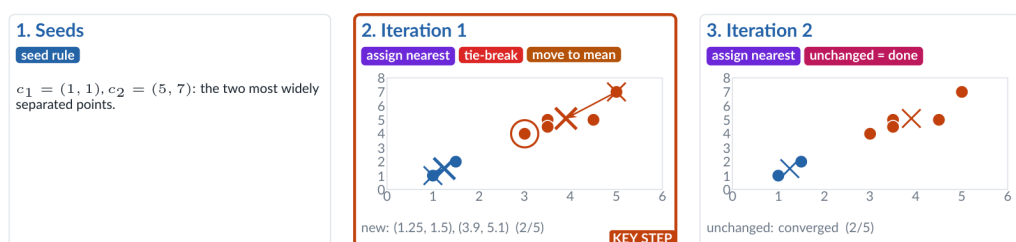
- (3, 4) is an exact tie: $\sqrt{4+9} = \sqrt{4+9} = 3.606$ both ways. **State a tie-break rule and use it** – here, send it to C2. Examiners give the mark for naming the rule, not for the direction chosen.
- C1 → centroid (1.25, 1.5)
- C2 → $\left(\frac{3+5+3.5+4.5+3.5}{5}, \frac{4+7+5+5+4.5}{5}\right) = (3.9, 5.1)$

- **Assignment unchanged → converged.** The tie at (3, 4) is now decisive (1.421 against 3.052), so the arbitrary choice in iteration 1 did not matter here.

Final answer

C1 = {(1, 1), (1.5, 2)} centroid (1.25, 1.5); C2 = {(3, 4), (5, 7), (3.5, 5), (4.5, 5), (3.5, 4.5)} centroid (3.9, 5.1).

Step by step: the exact tie at (3,4), ringed in orange, sent to C2



Problem 1.5 8

QUESTION AS PRINTED 2076 Chaitra · Q9 · [8]

Given the matrix X whose rows represent different data points, perform a k-means clustering on this dataset using the Euclidean distance as the distance function. Here K is chosen as 3. The center of the 3 clusters are initialized as red (6.2, 3.2), green (6.6, 3.7) and blue (6.5, 3.0). Provide the final cluster centers and comment on the number of iterations required for the clusters to converge.

$$X = \begin{bmatrix} 5.9 & 3.2 \\ 4.6 & 2.9 \\ 6.2 & 2.8 \\ 4.7 & 3.2 \\ 5.5 & 4.2 \\ 5.0 & 3.0 \\ 4.9 & 3.1 \\ 6.7 & 3.1 \\ 5.1 & 3.8 \\ 6.0 & 3.0 \end{bmatrix}$$

Iteration 1

(assign to nearest of the 3 given centres)

- **red** $\{p_1, p_2, p_4, p_6, p_7, p_9, p_{10}\}$ **green** $\{p_5\}$ **blue** $\{p_3, p_8\}$
- New centres: red (5.171, 3.171), green (5.5, 4.2), blue (6.45, 2.95)

Iteration 2

- **red** $\{p_2, p_4, p_6, p_7\}$ **green** $\{p_5, p_9\}$ **blue** $\{p_1, p_3, p_8, p_{10}\}$
- New centres: red (4.8, 3.05), green (5.3, 4.0), blue (6.2, 3.025)

Iteration 3

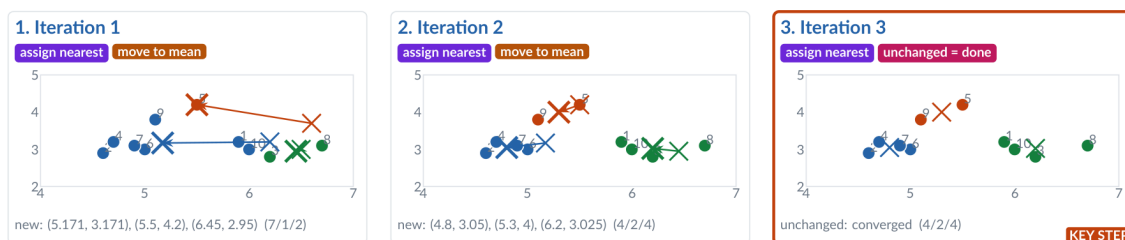
- Every point stays where it was → **converged**.

Final answer

Cluster	Members	Final centre
red	p_2, p_4, p_6, p_7	(4.80, 3.05)
green	p_5, p_9	(5.30, 4.00)
blue	p_1, p_3, p_8, p_{10}	(6.20, 3.025)

- **Comment on iterations:** assignments changed at iterations 1 and 2 and were stable at iteration 3, so **convergence took 3 iterations (2 real updates + 1 confirming pass)**.
- Worth adding: the 3 supplied centres were all bunched on the right of the data ($x \approx 6.2-6.6$), yet K-means still pulled them apart into a sensible left / middle / right split. Good illustration that it converges quickly even from a poor start.

Step by step: red, green, blue centres (blue, orange, green here) over three passes



Problem 1.6 – the defective question 5+3

QUESTION AS PRINTED 2081 Baishakh · Q4 (b) · [5+3]

Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm.

(nothing follows on the paper – the next line is Q5)

The paper prints **NO data table for this question** — it jumps straight from the sentence to Q5. Verified against the original scan and recorded in the PYQ archive.

In the exam: answer the 3-mark demerits part in full (\$5.2), write the 5 steps of the algorithm, and demonstrate on any small dataset of your own, stating that you assumed one. Use **Problem 1.1's** 6 points — it is the dataset this paper's sister questions use. **verified/added**

Problem 1.7 2+8

QUESTION AS PRINTED 2082 Bhadra · Q6 · [2+8]

Describe the K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering.

Instance	A	B
1	1	2
2	1.5	1
3	3.5	1.5
4	4	3
5	3.5	2.5
6	6	4

Seeds

the two most widely separated points, $c_1 = (1, 2)$ and $c_2 = (6, 4)$.

Iteration 1

Point	d to $(1, 2)$	d to $(6, 4)$	Cl.
(1, 2)	0	5.385	C1
(1.5, 1)	1.118	5.408	C1
(3.5, 1.5)	2.550	3.536	C1
(4, 3)	3.162	2.236	C2
(3.5, 2.5)	2.550	2.915	C1
(6, 4)	5.385	0	C2
- C1 $\rightarrow \left(\frac{1+1.5+3.5+3.5}{4}, \frac{2+1+1.5+2.5}{4} \right) = (2.375, 1.75)$			
- C2 $\rightarrow \left(\frac{4+6}{2}, \frac{3+4}{2} \right) = (5, 3.5)$			

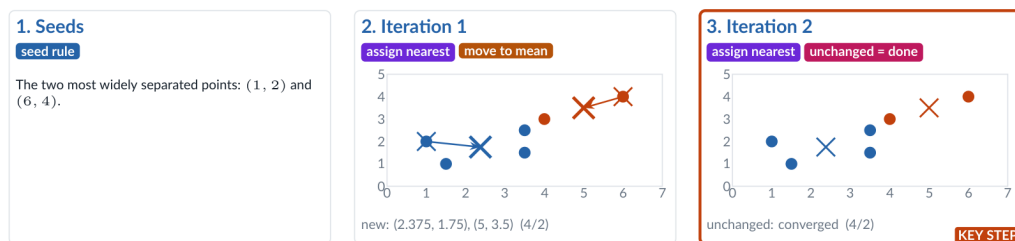
Iteration 2

Point	d to $(2.375, 1.75)$	d to $(5, 3.5)$	Cl.
(1, 2)	1.398	4.272	C1
(1.5, 1)	1.152	4.301	C1
(3.5, 1.5)	1.152	2.500	C1
(4, 3)	2.050	1.118	C2
(3.5, 2.5)	1.352	1.803	C1
(6, 4)	4.267	1.118	C2
- Assignment unchanged $\rightarrow$ converged after 2 iterations.			

Final answer

- **Cluster 1** = $\{(1, 2), (1.5, 1), (3.5, 1.5), (3.5, 2.5)\}$, centroid $(2.375, 1.75)$
- **Cluster 2** = $\{(4, 3), (6, 4)\}$, centroid $(5, 3.5)$

Step by step: seeds, then each iteration



Problem 1.8 5+3

QUESTION AS PRINTED 2081 Bhadra · Q8 · [5+3]

Consider the data points $\{(2, 3), (3, 3), (6, 8), (8, 8), (7, 5)\}$ with initial centroid $(2, 3)$. Apply K-means algorithm to choose the next centroid and cluster these data points into $k = 2$. Discuss the problems that arise with K-means clustering algorithm and ways to solve it.

Only **ONE centroid is given**, and "choose the next centroid" is the first thing asked. Take the **farthest point** from the given one — the standard max-separation seeding rule — and say so.

Step 1: choose the second centroid

Distance from (2, 3)	(3, 3)	(6, 8)	(8, 8)	(7, 5)
Euclidean	1.000	6.403	7.810	5.385

(8, 8) is farthest $\rightarrow c_2 = (8, 8)$.

Iteration 1

Point	d to (2, 3)	d to (8, 8)	Cl.
(2, 3)	0	7.810	C1
(3, 3)	1.000	7.071	C1
(6, 8)	6.403	2.000	C2
(8, 8)	7.810	0	C2
(7, 5)	5.385	3.162	C2

- C1 $\rightarrow (2.5, 3)$ C2 $\rightarrow (\frac{6+8+7}{3}, \frac{8+8+5}{3}) = (7, 7)$

Iteration 2

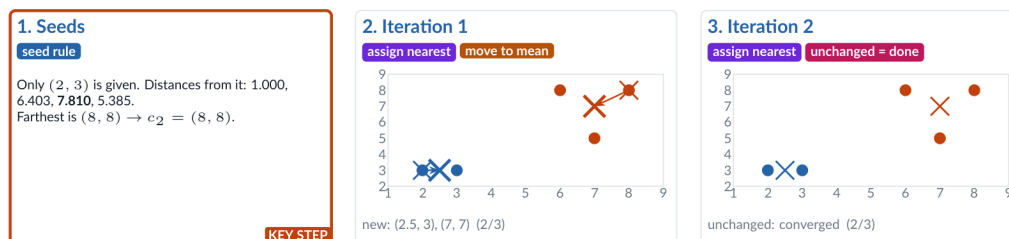
Point	d to (2.5, 3)	d to (7, 7)	Cl.
(2, 3)	0.500	6.403	C1
(3, 3)	0.500	5.657	C1
(6, 8)	6.103	1.414	C2
(8, 8)	7.433	1.414	C2
(7, 5)	4.924	2.000	C2

- **Unchanged $\rightarrow$ converged.**

Final answer

C1 = {(2, 3), (3, 3)} centroid (2.5, 3); C2 = {(6, 8), (8, 8), (7, 5)} centroid (7, 7). The 3-mark “problems and ways to solve” half is the limitations list in §5.2.

Step by step: choose the second seed, then iterate



Problem 1.9 4+8

QUESTION AS PRINTED 2073 Chaitra · Q7 · [4+8]

How clustering differ from classification? Given the one-dimensional points {5, 12, 18, 24, 30, 42, 48} with initial centroids {5, 12, 18}, create three clusters by K-Means algorithm and calculate SSE for this clustering result.

One dimension, so “distance” is just $|x - c|$. This is the only PYQ that asks for SSE, and it is the only one that runs past two iterations.

Iter.	Assignment (nearest centroid)	Clusters	New centroids
1	centroids 5, 12, 18	{5}, {12}, {18, 24, 30, 42, 48}	5, 12, $\frac{162}{5} = 32.4$
2	centroids 5, 12, 32.4	{5}, {12, 18}, {24, 30, 42, 48}	5, 15, $\frac{144}{4} = 36$
3	centroids 5, 15, 36	{5}, {12, 18, 24}, {30, 42, 48}	5, 18, 40
4	centroids 5, 18, 40	{5}, {12, 18, 24}, {30, 42, 48}	5, 18, 40 unchanged

- **Converged at iteration 4.** Note the cluster boundaries kept moving for three rounds — do not stop after two just because the earlier problems did.

SSE

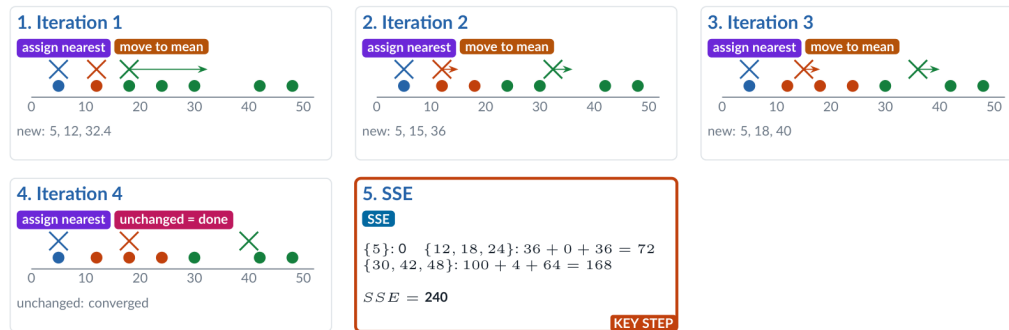
$$SSE = \sum_i \sum_{x \in C_i} (x - c_i)^2$$

Cluster	Centroid	Squared errors	Subtotal
{5}	5	$(5 - 5)^2 = 0$	0
{12, 18, 24}	18	$(12 - 18)^2 + (18 - 18)^2 + (24 - 18)^2 = 36 + 0 + 36$	72
{30, 42, 48}	40	$(30 - 40)^2 + (42 - 40)^2 + (48 - 40)^2 = 100 + 4 + 64$	168

$$SSE = 0 + 72 + 168 = 240$$

- **SSE is the objective K-means minimises**, so quoting it is how you show the clustering is the algorithm’s answer and not just a guess. A different seeding that gave a larger SSE would be a worse local optimum.

Step by step: four passes along the line (the boundaries move three times), then SSE



Problem 1.10 – the Manhattan one 8

QUESTION AS PRINTED 2079 Baishakh · Q8 · [8]

Define clustering. Given the matrix whose column represent different data points, perform a K-means clustering on this dataset using the **Manhattan** as the distance function. The center of the 3 clusters are initiated as $A(6.2, 3.2)$, $B(6.6, 3.7)$ and $C(6.5, 3.0)$. Provide the final cluster centers after 3 iteration.

x	5.9	4.6	6.2	4.7	5.5	5.0	4.9	6.7	5.1	6.0
y	3.2	2.9	2.8	3.2	4.2	3.0	3.1	3.1	3.8	3.0

Manhattan, not Euclidean: $d = |x_1 - x_2| + |y_1 - y_2|$, no squares and no square root. **Stop after 3 iterations** – the question says so, and it has *not* converged by then. Say that in the answer.

Iteration 1

centres $A(6.2, 3.2)$, $B(6.6, 3.7)$, $C(6.5, 3.0)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.30	1.20	0.80	A
(4.6, 2.9)	1.90	2.80	2.00	A
(6.2, 2.8)	0.40	1.30	0.50	A
(4.7, 3.2)	1.50	2.40	2.00	A
(5.5, 4.2)	1.70	1.60	2.20	B
(5.0, 3.0)	1.40	2.30	1.50	A
(4.9, 3.1)	1.40	2.30	1.70	A
(6.7, 3.1)	0.60	0.70	0.30	C
(5.1, 3.8)	1.70	1.60	2.20	B
(6.0, 3.0)	0.40	1.30	0.50	A

New centres: $A = (5.329, 3.029)$ (7 points), $B = (5.3, 4.0)$ (2), $C = (6.7, 3.1)$ (1).

Iteration 2

centres $A(5.329, 3.029)$, $B(5.3, 4.0)$, $C(6.7, 3.1)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.74	1.40	0.90	A
(4.6, 2.9)	0.86	1.80	2.30	A
(6.2, 2.8)	1.10	2.10	0.80	C
(4.7, 3.2)	0.80	1.40	2.10	A
(5.5, 4.2)	1.34	0.40	2.30	B
(5.0, 3.0)	0.36	1.30	1.80	A
(4.9, 3.1)	0.50	1.30	1.80	A
(6.7, 3.1)	1.44	2.30	0.00	C
(5.1, 3.8)	1.00	0.40	2.30	B
(6.0, 3.0)	0.70	1.70	0.80	A

New centres: $A = (5.183, 3.067)$ (6), $B = (5.3, 4.0)$ (2), $C = (6.45, 2.95)$ (2).

Iteration 3

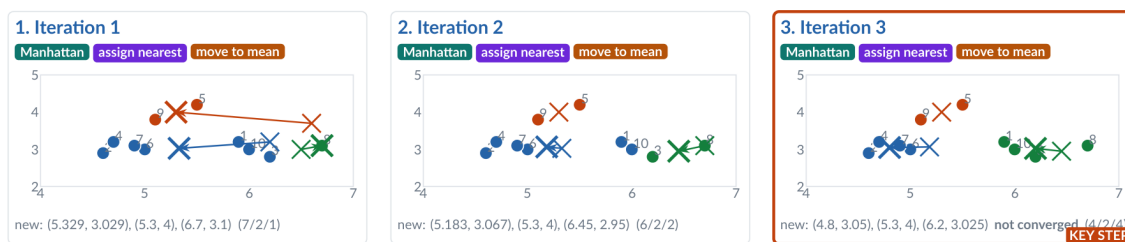
centres $A(5.183, 3.067)$, $B(5.3, 4.0)$, $C(6.45, 2.95)$

Point	d to A	d to B	d to C	Cluster
(5.9, 3.2)	0.85	1.40	0.80	C
(4.6, 2.9)	0.75	1.80	1.90	A
(6.2, 2.8)	1.28	2.10	0.40	C
(4.7, 3.2)	0.62	1.40	2.00	A
(5.5, 4.2)	1.45	0.40	2.20	B
(5.0, 3.0)	0.25	1.30	1.50	A
(4.9, 3.1)	0.32	1.30	1.70	A
(6.7, 3.1)	1.55	2.30	0.40	C
(5.1, 3.8)	0.82	0.40	2.20	B
(6.0, 3.0)	0.88	1.70	0.50	C

Final answer after 3 iterations

- $A = (4.8, 3.05)$ from $\{(4.6, 2.9), (4.7, 3.2), (5.0, 3.0), (4.9, 3.1)\}$
- $B = (5.3, 4.0)$ from $\{(5.5, 4.2), (5.1, 3.8)\}$
- $C = (6.2, 3.025)$ from $\{(5.9, 3.2), (6.2, 2.8), (6.7, 3.1), (6.0, 3.0)\}$
- **The assignment was still changing at iteration 3**, so this is *not* a converged clustering. The question asked for 3 iterations, so stop there and **state that it has not converged** – that sentence is worth a mark.
- **Note how the three seeds were all bunched near (6.5, 3)**, which is why cluster C started with a single point and had to grow every round. A textbook illustration of **poor seeding**.

Step by step: Manhattan distance, three passes, still moving



M2. Agglomerative hierarchical clustering

Method

1. Start with each point as its own cluster; write the distance matrix.
 2. Find the **smallest** entry → merge those 2 clusters. **Record the merge height** – that value is the dendrogram bar.
 3. Update the matrix using the linkage rule: **single** = MIN of the two rows, **complete** = MAX.
 4. Repeat until one cluster remains, then draw the dendrogram with merges at their heights.
- **Merge heights never decrease.** If yours does, you have made an arithmetic slip.
 - n points always give exactly $n - 1$ merges.

Problem 2.1 2+7

QUESTION AS PRINTED 2080 Bhadra · Q4 (b) · [2+7]

What is hierarchical clustering? Use this clustering approach to draw dendrogram for given data points.

	p_1	p_2	p_3	p_4	p_5	p_6
p_1	0.00	0.24	0.22	0.37	0.34	0.23
p_2	0.24	0.00	0.15	0.20	0.14	0.25
p_3	0.22	0.15	0.00	0.15	0.28	0.11
p_4	0.37	0.20	0.15	0.00	0.29	0.22
p_5	0.34	0.14	0.28	0.29	0.00	0.39
p_6	0.23	0.25	0.11	0.22	0.39	0.00

Distances are given directly, so no coordinate work. The paper does not name a linkage — use **single link (MIN)** and say so. **Recompute and rewrite the whole matrix after every merge** — that redrawing is the method, and it is where the marks are.

Step 1

smallest entry in the matrix above is $d(p_3, p_6) = 0.11 \rightarrow$ merge at height **0.11**

Matrix after merge 1

5 clusters

	p_1	p_2	p_4	p_5
p_1	0.00	0.24	0.37	0.34
p_2	0.24	0.00	0.20	0.14
p_4	0.37	0.20	0.00	0.29
p_5	0.34	0.14	0.29	0.00
$\{p_3p_6\}$	0.22	0.15	0.15	0.28

How the new row was computed

MIN of the two old rows

- $d(\{p_3p_6\}, p_1) = \min(0.22, 0.23) = 0.22$
- $d(\{p_3p_6\}, p_2) = \min(0.15, 0.25) = 0.15$
- $d(\{p_3p_6\}, p_4) = \min(0.15, 0.22) = 0.15$
- $d(\{p_3p_6\}, p_5) = \min(0.28, 0.39) = 0.28$
- Every other entry is copied over unchanged.

Step 2

smallest is now $d(p_2, p_5) = 0.14 \rightarrow$ merge at height **0.14**

Matrix after merge 2

4 clusters

	p_1	p_4	$\{p_3p_6\}$
p_1	0.00	0.37	0.22
p_4	0.37	0.00	0.15
$\{p_3p_6\}$	0.22	0.15	0.00
$\{p_2p_5\}$	0.24	0.20	0.15

New row

- $d(\{p_2p_5\}, p_1) = \min(0.24, 0.34) = 0.24$
- $d(\{p_2p_5\}, p_4) = \min(0.20, 0.29) = 0.20$
- $d(\{p_2p_5\}, \{p_3p_6\}) = \min(0.15, 0.28) = 0.15$
- **A tie at 0.15**, between $(\{p_3p_6\}, p_4)$ and $(\{p_3p_6\}, \{p_2p_5\})$. **Either may be merged first** and the dendrogram heights come out the same; state the rule you use. Taken here in the order the matrix is read.

Step 3

take $d(\{p_3p_6\}, p_4) = 0.15 \rightarrow$ merge at height **0.15**

Matrix after merge 3

3 clusters

	p_1	$\{p_2p_5\}$
p_1	0.00	0.24
$\{p_2p_5\}$	0.24	0.00
$\{p_3p_4p_6\}$	0.22	0.15

New row

- $d(\{p_3p_4p_6\}, p_1) = \min(0.22, 0.37) = 0.22$
- $d(\{p_3p_4p_6\}, \{p_2p_5\}) = \min(0.15, 0.20) = 0.15$

Step 4

smallest is **0.15** $\rightarrow$ merge $\{p_3p_4p_6\}$ with $\{p_2p_5\}$ at height **0.15**

Matrix after merge 4

2 clusters

	p_1
p_1	0.00
$\{p_2p_3p_4p_5p_6\}$	0.22

New row

- $d(\{p_2p_3p_4p_5p_6\}, p_1) = \min(0.24, 0.22) = 0.22$, the smaller of the two surviving links to p_1 .

Step 5

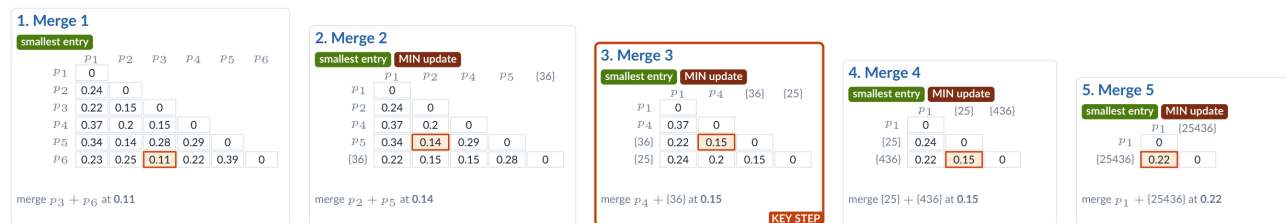
only one distance left $\rightarrow$ merge everything at height **0.22**

Final answer

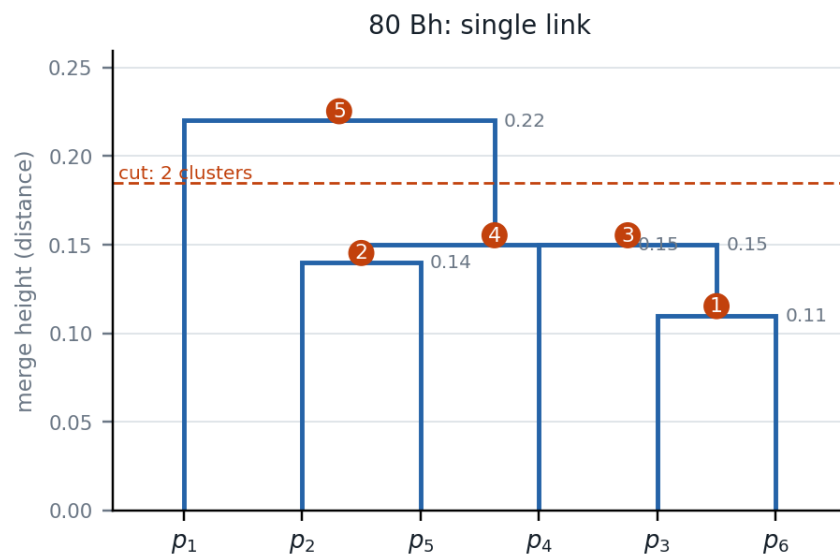
Dendrogram heights: **0.11, 0.14, 0.15, 0.15, 0.22**.

- Reading it: $\{p_3, p_6\}$ is the tightest pair; p_1 is the most isolated point, joining last.
- Cutting between 0.15 and 0.22 gives **2 clusters**: $\{p_1\}$ and $\{p_2, p_3, p_4, p_5, p_6\}$.

Step by step: the matrix before each merge, smallest entry lit (single link)



The dendrogram



Bars at the five merge heights, numbered in merge order (Steps 1 to 5 above); merges 3 and 4 share the height 0.15, the tie. The dashed line is the 2-cluster cut. Drawn by `dendfig.py`, which replays the merges and checks them against this page.

Problem 2.2 8

QUESTION AS PRINTED 2080 Baishakh · Q7 · [8]

Cluster the following samples based on complete-linkage algorithm and draw the dendrogram. Using Euclidean distance.

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

Step 1: Euclidean distance matrix

$p_1(0.40, 0.53)$ $p_2(0.22, 0.38)$ $p_3(0.35, 0.32)$ $p_4(0.26, 0.19)$ $p_5(0.08, 0.41)$ $p_6(0.45, 0.30)$

	p_1	p_2	p_3	p_4	p_5
p_2	0.234				
p_3	0.216	0.143			
p_4	0.368	0.194	0.158		
p_5	0.342	0.143	0.285	0.284	
p_6	0.235	0.244	0.102	0.219	0.386

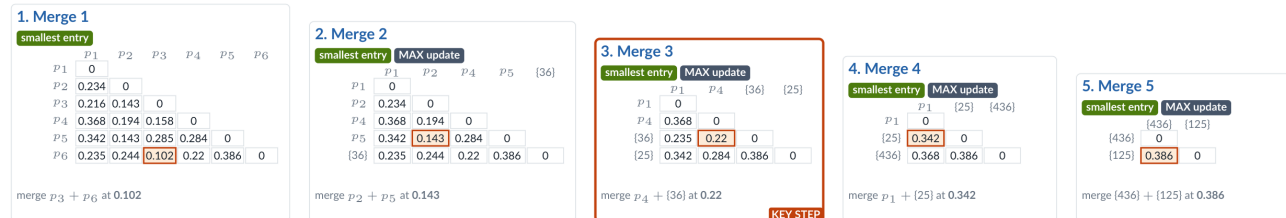
Step 2: merge sequence (MAX update)

Step	Merge	Height	Updated distances (MAX)
1	p_3, p_6	0.102	to p_1 0.235, p_2 0.244, p_4 0.219, p_5 0.386
2	p_2, p_5	0.143	to p_1 0.342, p_4 0.284, $\{p_3p_6\}$ 0.386
3	$\{p_3p_6\}, p_4$	0.219	to p_1 0.368, $\{p_2p_5\}$ 0.386
4	$p_1, \{p_2p_5\}$	0.342	to $\{p_3p_4p_6\}$ 0.386
5	the two remaining	0.386	—

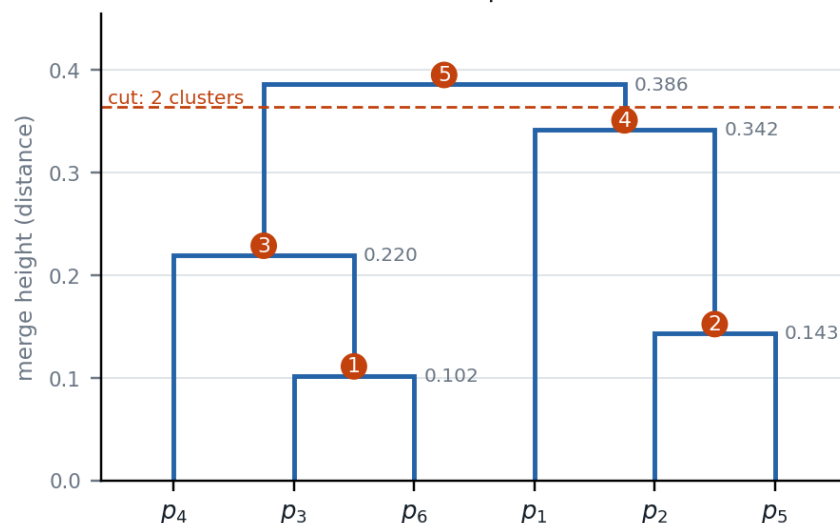
Final answer

Heights **0.102, 0.143, 0.219, 0.342, 0.386**; cutting below 0.386 gives $\{p_1, p_2, p_5\}$ and $\{p_3, p_4, p_6\}$.

Step by step: the matrix before each merge (complete link: the merged row takes the MAX)



80 Ba: complete link



The dendrogram the question asks for, merges numbered as in the Step 2 table; the cut below 0.386 leaves the two groups. From `dendfig.py`.

Problem 2.3 4+4

QUESTION AS PRINTED 2076 Chaitra · Q10 · [4+4]

The table below is a distance matrix for six objects:

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single-link by drawing a dendrogram.
- Show the final result of hierarchical clustering with complete-link by drawing a dendrogram.

The ideal question for showing you understand the difference — one matrix, both linkages.

(a) Single link (MIN)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, {CD}	0.16
4	{ABCD}, F	0.20
5	all, + E	0.28

- Heights: **0.12, 0.14, 0.16, 0.20, 0.28**
- Everything merges early and at similar heights → **chaining**.

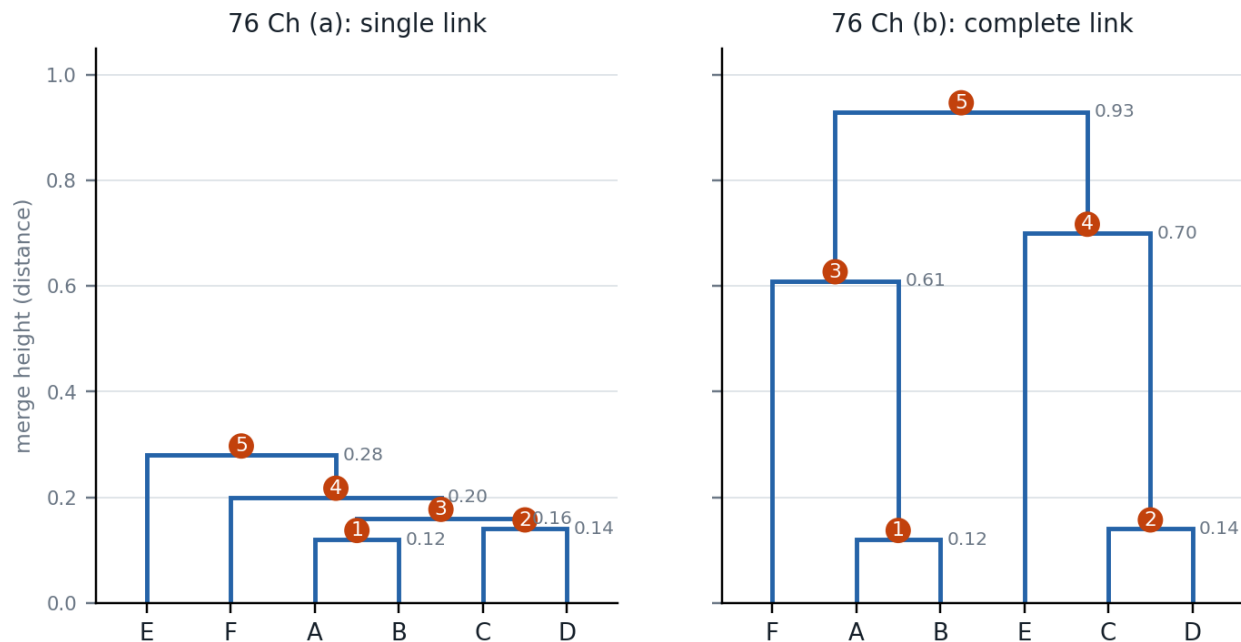
(b) Complete link (MAX)

	Merge	Height
1	A, B	0.12
2	C, D	0.14
3	{AB}, F	0.61
4	{CD}, E	0.70
5	the two groups	0.93

- Heights: **0.12, 0.14, 0.61, 0.70, 0.93**
- Final 2 clusters: **{A,B,F}** and **{C,D,E}** – a balanced split.

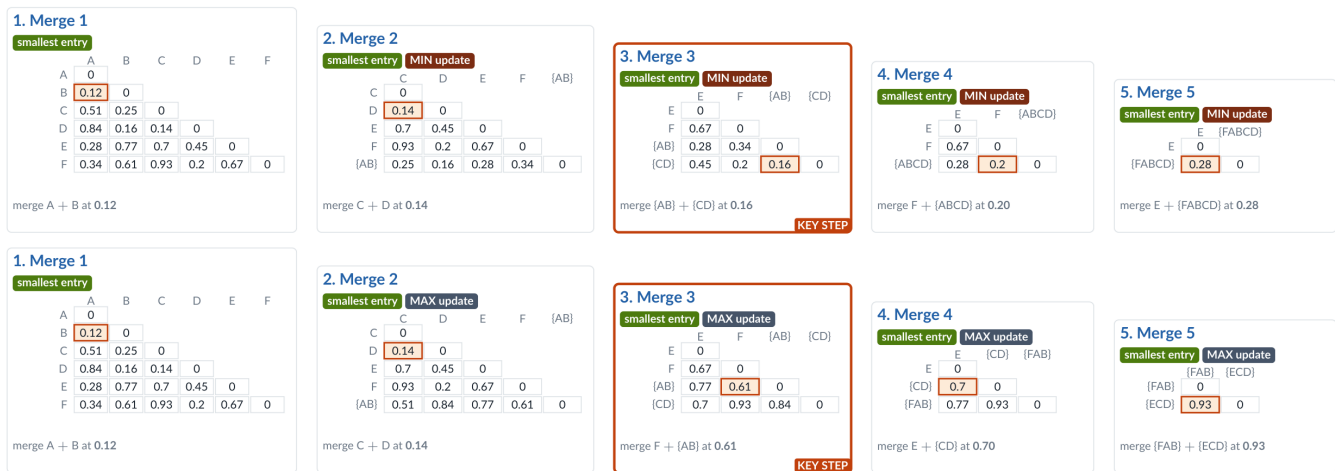
The comparison is the marks. The first 2 merges are identical, then the methods diverge completely: single link strings everything onto the A-B-C-D chain and leaves E as a straggler, while complete link produces two clean balanced groups. **Single link** chases the nearest neighbour, so it follows chains and elongated shapes. **Complete link** is held back by the worst pair, so it prefers compact globular clusters. verified/added

Both dendrograms, to one scale



(a) single link: everything joins below 0.3, the chaining in one look. (b) complete link: two compact groups that only meet at 0.93. Merges numbered as in the two tables. From `dendfig.py`.

Step by step: (a) single link, top row; (b) complete link, bottom row



M3. DBSCAN point classification

Method

- For each point, count how many points lie within ε (include the point itself).
- Count $\geq \text{MinPts} \rightarrow$ **core**.
- Otherwise, within ε of a core point $\rightarrow$ **border**.
- Otherwise $\rightarrow$ **noise**.
- Join core points within ε of each other into clusters; attach the border points.

Problem 3.1 3+5

QUESTION AS PRINTED 2082 Bhadra · Q5 · [3+5]

What are core, border and noise points? Consider the following points (2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4), (1, 2), (4, 9). Use DBSCAN algorithm to find cluster with $\varepsilon = 2$ and $\text{minPts} = 2$.

The paper prints the radius as a plain e ; it is ε . Label the points $A \dots H$ in the order given, so the working can name them.

Step 1: neighbourhood of every point

$\varepsilon = 2$, and **the point counts itself**, so $\text{MinPts} = 2$ means “at least one other point within 2”.

Point	Others within $\varepsilon = 2$	$ N $	Type	Why
A(2, 10)	— (nearest is H at 2.236)	1	noise	no neighbour
B(2, 5)	— (nearest is G at 3.162)	1	noise	no neighbour
C(8, 4)	E (1.414), F (2.000)	3	core	≥ 2
D(5, 8)	H (1.414)	2	core	≥ 2
E(7, 5)	C (1.414), F (1.414)	3	core	≥ 2
F(6, 4)	C (2.000), E (1.414)	3	core	≥ 2
G(1, 2)	— (nearest is B at 3.162)	1	noise	no neighbour
H(4, 9)	D (1.414)	2	core	≥ 2

- C–F is exactly 2.000**, which is **inside** the neighbourhood: the definition is $d \leq \varepsilon$, not $d < \varepsilon$. Getting this wrong splits the cluster.
- There are no border points here.** A, B and G are not core, and none of them is within ε of a core point either, so all three are noise. Say that explicitly – the question names all three categories.

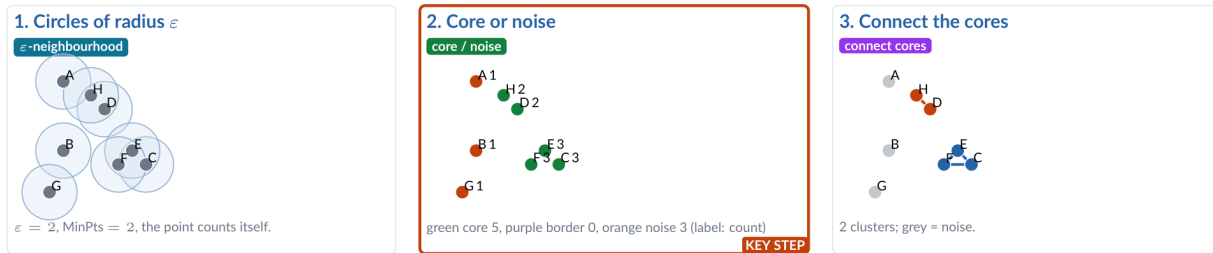
Step 2: join the core points

- C, E, F are mutually within $\varepsilon \rightarrow$ **Cluster 1** = {(8, 4), (7, 5), (6, 4)}.
- D and H are within ε of each other and of nothing else $\rightarrow$ **Cluster 2** = {(5, 8), (4, 9)}.
- No border point is left to attach.

Final answer

- **2 clusters:** $\{(8, 4), (7, 5), (6, 4)\}$ and $\{(5, 8), (4, 9)\}$.
- **Core:** all 5 of those points. **Border:** none. **Noise:** $(2, 10), (2, 5), (1, 2)$.
- **Check:** $5 + 0 + 3 = 8$ points. ✓

Step by step: neighbourhoods, point types, clusters



Practice – the same method with MinPts = 3

PROBLEM not a PYQ · written to show the MinPts effect **verified/added**

Classify each of the points $A(1, 1)$, $B(1, 2)$, $C(2, 1)$, $D(8, 8)$, $E(8, 9)$, $F(25, 25)$ as a **core**, **border** or **noise** point under DBSCAN with $\varepsilon = 2$ and MinPts = 3 (Euclidean distance, the point itself counted in its own neighbourhood). Then give the clusters DBSCAN produces.

Kept because it isolates **what MinPts alone changes**: the same shaped data gives a different answer at MinPts 3 than at MinPts 2. **verified/added**

Step 1: neighbourhood counts

(including the point itself)

Point	Points within $\varepsilon = 2$	Count	Type
$A(1, 1)$	A, B (1.00), C (1.00)	3	core
$B(1, 2)$	B, A (1.00), C (1.41)	3	core
$C(2, 1)$	C, A (1.00), B (1.41)	3	core
$D(8, 8)$	D, E (1.00)	2	not core
$E(8, 9)$	E, D (1.00)	2	not core
$F(25, 25)$	F only	1	noise

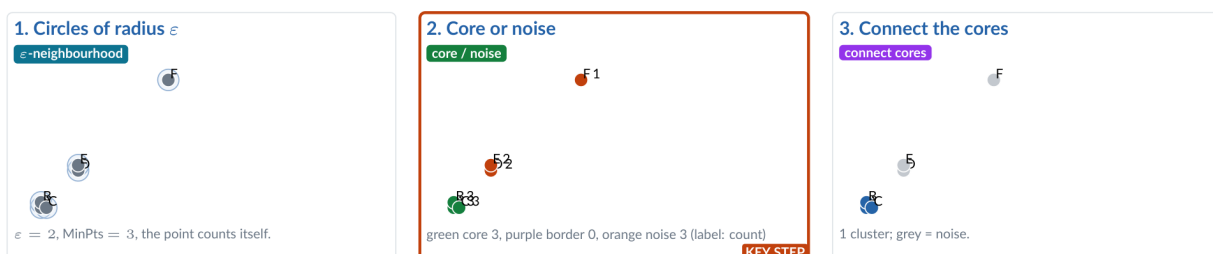
Step 2: form clusters

- A, B, C are all core and all within ε of each other → **Cluster 1** = $\{A, B, C\}$.
- D and E have only 2 neighbours each, and **neither is within ε of any core point** → they are **not border points** either, they are **noise**.
- F is isolated → **noise**.

Final answer

- **1 cluster:** $\{A, B, C\}$. **Noise:** D, E, F .
- **The trap:** D and E look like a cluster to the eye, but with MinPts = 3 a pair can never form one. **A border point must be within ε of an actual core point** – being near a non-core point is not enough.
- Lower MinPts to 2 and $\{D, E\}$ becomes a second cluster. **Quote your parameters with the answer.**

Step by step: the same method with MinPts 3 (D and E are noise)



Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.